MAJOR PROJECT TECHNICAL ABSTRACT

System Design with AI: Automated Architectural Analysis, Rule Validation & Interview Simulation Platform

An AI-Augmented Engineering Platform Combining Deterministic Graph Traversal, Knowledge RAG, and Asynchronous Message Queues

Domain	Distributed Systems, Applied AI & Software Engineering	**Architecture**	Modular Monolith (BFF) + Event-Driven Worker Queue
Tech Stack	React Flow, TypeScript, NestJS, PostgreSQL, Redis, Redpanda	**AI / RAG**	pgvector Cosine Search, Gemini/Groq LLMs, Tool Calling

PROJECT ABSTRACT

Designing scalable, resilient distributed systems is a fundamental challenge in software engineering, yet traditional interview preparation lacks automated, objective feedback. Moreover, relying solely on large language models often results in architectural hallucinations and erroneous graph calculations. To address this, this project presents an AI-augmented system design studio that couples deterministic graph algorithms with retrieval-augmented generation (RAG) and asynchronous processing. Built using React Flow, TypeScript, NestJS, PostgreSQL with pgvector, Redis, and Redpanda, the platform implements an interactive architectural canvas decoupled from UI pixel coordinates via a domain normalizer. The evaluation pipeline adheres to a strict deterministic-before-AI execution boundary: a graph metrics engine first detects circular dependencies via depth-first search (DFS), calculates fan-out, and identifies single points of failure (SPOF). Next, an extensible rule engine flags critical anti-patterns in under a millisecond. Contextual vectors from a curated distributed systems knowledge base are then retrieved via cosine similarity and injected into the LLM prompt alongside dynamic tool-calling interfaces. Heavy evaluation workloads are offloaded to Kafka-compatible Redpanda message queues with background workers and real-time WebSocket progress streaming. The expected outcome is an objective, high-performance platform capable of providing sub-second deterministic validation and citation-grounded architectural critiques. By eliminating AI hallucinations and bridging the gap between theoretical system design and rigorous architectural standards, the project significantly enhances candidate preparation and distributed system modeling.

Keywords: *System Design, Distributed Systems, Graph Traversal (DFS), Single Point of Failure (SPOF), Deterministic Rule Engine, Retrieval-Augmented Generation (RAG), pgvector, Asynchronous Message Queues (Redpanda), WebSocket Streaming.*

KEY ARCHITECTURAL HIGHLIGHTS & METHODOLOGY

1. Deterministic Graph Math (Pre-AI)
DFS cycle detection, component out-degree fan-out, and SPOF identification run mathematically in <1ms before LLM invocation, eliminating structural hallucinations.

2. Standalone Rule Engine
Sub-millisecond validation checks flag architectural violations: unprotected client database access, unreplicated storage, missing cache tiers, and synchronous coupling.

3. Grounded Knowledge Base RAG
15 authoritative distributed systems patterns embedded and retrieved via cosine similarity over pgvector, allowing the AI critique to cite exact industry pattern codes.

4. Asynchronous Queue & WebSockets
Redpanda pub/sub broker decouples evaluation workloads from synchronous HTTP threads, with workers streaming progressive pipeline status updates via WebSockets.